

Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Author: Not present in source JSON

Executive Summary

- The BRD outlines requirements for ingesting customer and order CSV data into Delta tables, applying cleansing, computing TotalAmount, joining for ordersummary with SCD Type 2 history, and aggregating daily spend in customeraggregatespend, all implemented via Delta Live Tables.
- System Name: Application

Business Objectives

- Load source CSVs into Delta tables for customer and order data.
- Cleanse data by removing "Null"/Null literals and duplicates.
- Enrich orders with a computed TotalAmount as PricePerUnit × Qty.
- Create an ordersummary table by joining customer and order on CustId.
- Apply SCD Type 2 processing for customer data to maintain historical records.
- Aggregate spend per Name and Date in customeraggregatespend.
- Execute all steps within Delta Live Tables.

In-Scope

- Source ingestion from specified file system locations.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Defined cleansing actions.
- Computation of TotalAmount.
- Join and SCD Type 2 logic for ordersummary.
- Aggregate processing for customeraggregatespend.
- Implementation through Delta Live Tables.

Out-of-Scope

- Data sources, systems, or fields not in source JSON.
- Security, access controls, SLAs, monitoring, alerting.
- Performance tuning, scaling parameters.
- Scheduling, orchestration, or DevOps beyond "implement steps 1–10 using Delta Live Tables".

- Data quality metrics beyond stated cleansing.
- User interfaces, reports, or analytics layers.

Stakeholders

- Not present in source JSON

Business Requirements

BR-CORE-001 — Source Ingestion

- Description: Load customer and order CSVs from defined paths into Delta tables.
- Business Value: Establishes foundational datasets for downstream processing.
- Priority: High

BR-CORE-002 — Schema Definition

- Description: Apply explicit schemas for customer and order tables as indicated in requirements.
- Business Value: Ensures consistent structure for all processes.
- Priority: High

BR-CORE-003 — Computed Columns

- Description: Add and calculate TotalAmount column in order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Automates order enrichment for accurate financial data.
- Priority: High

BR-CORE-004 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from both customer and order tables.
- Business Value: Improves data quality for reliable processing.
- Priority: High

BR-CORE-005 — Ordersummary Creation

- Description: Create the ordersummary table by joining customer and order tables on CustId, applying SCD Type 2 for customer attributes.
- Business Value: Provides a historical, cleansed, and joined dataset for analysis.
- Priority: High

BR-CORE-006 — SCD Type 2 Handling

- Description: Set previous customer records to Inactive, new records to Active, and update StartDate and EndDate in ordersummary to track changes over time.
- Business Value: Maintains audit-ready version history for compliance and analytics.

- Priority: High

BR-CORE-007 — Spend Aggregation

- Description: Aggregate spend for each customer by Name and Date in customeraggregatespend.
- Business Value: Enables insights into customer purchasing trends.
- Priority: Medium

BR-CORE-008 — Delta Live Tables Implementation

- Description: Orchestrate the pipeline steps using Delta Live Tables.
- Business Value: Utilizes automation for reliability and maintainability.
- Priority: High

Assumptions & Constraints

- No assumptions beyond explicit requirements.
- Implementation must utilize Delta Live Tables.
- Attributes for SCD Type 2 (StartDate, EndDate, Active) referenced but not fully specified.
- All table and column naming strictly follows source JSON.

Success Metrics

- Accurate and complete ingestion of customer and order CSV data.
- No "Null" or duplicate records in managed tables.
- Correct computation of TotalAmount field.
- Full SCD Type 2 history for customer changes in ordersummary.
- Accurate daily spend aggregates in customeraggregatespend.
- All steps automated under Delta Live Tables.

Risks & Mitigations

- Ambiguous SCD Type 2 attribute definitions (risk: misimplementation)
- Mitigation: Flag for clarification with stakeholders.
- Data quality issues beyond literal cleansing (risk: undetected anomalies)
- Mitigation: Recommend augmenting with additional quality checks.
- Monitoring and alerting not defined (risk: pipeline failures unnoticed)
- Mitigation: Suggest implementation-specific alerting in future phases.

Approval

- Approval details not present in source JSON.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	customer	CustId, Name, EmailId, Region	Target for customer
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Target for order s
y	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName,	Join of customer a
Qty, Date	SCD Type 2 history			
regatespend	Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregation on or

Appendix C: Transformations

	Transformation	Input	Output	Logic
puted column	TotalAmount = PricePerUnit × Qty	order	order	Expression-based
literals & duplicate	Cleansing	customer	customer	Remove "Null"/Null, deduplicate
literals & duplicate	Cleansing	order	order	Remove "Null"/Null, deduplicate
	Join	customer, order	ordersummary	Join on CustId
oe 2	Change history	ordersummary	ordersummary	Set Inactive/Active, StartDate, En
tion	Sum TotalAmount by Name, Date	ordersummary	customeraggregatespend	group by Name, Date, sum(Total

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-001	Computation	TotalAmount = PricePerUnit × Qty	Requirements
BR-002	Cleansing	Remove "Null"/Nulls, remove duplicates in customer and order	Requirements
BR-003	Join	Join on CustId for customer and order	Requirements
BR-004	SCD Type 2	Handle customer changes with SCD Type 2 in ordersummary	Requirements
BR-005	Aggregation	Sum TotalAmount by Name and Date in customeraggregatespend	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer CSV as customer table	Source CSV available
2	Ingest order CSV as order table	Source CSV available

3	Add computed column TotalAmount to order	Order ingestion
4	Cleanse customer and order tables	Prior load
5	Create ordersummary with join and SCD Type 2 processing	Cleansed tables
6	Aggregate spend to customeraggregatespend	ordersummary built

Appendix F: Data Pipeline Diagram

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